Project Report on

"Ladder diagram for an Elevator System"

Submitted
In Partial Fulfilment of the Requirements
For the Degree of

B. Tech. in Robotics and Automation

For the Subject

PROGAMMABLE LOGIC CONTROLLER AND HUMAN MACHINE INTERFACE

Submitted by

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CERTIFICATE

This is to certify that,

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have carried out the project entitled "**Ladder diagram for an Elevator System"** under my guidance in partial fulfilment of the requirement for the degree of Bachelor of Technology in Robotics and Automation of Bharati Vidyapeeth (Deemed to be University), Pune during the academic year 2025-2026 for the subject **PROGAMMABLE LOGIC CONTROLLER AND HUMAN MACHINE INTERFACE** for Project Based Learning (PBL) in the of Semester V.

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DECLARATION

We, hereby declare that the project titled **"Ladder diagram for an Elevator System"** being submitted by us towards the partial fulfilment of Bachelor of Technology, is a project-based learning work carried by us is our own work.

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ABSTRACT

An elevator system is an essential part of modern infrastructure, providing convenient vertical transportation in multi-story buildings. This project focuses on developing a **ladder logic diagram** for controlling an elevator system using a **Programmable Logic Controller (PLC)**. The system aims to ensure safe, efficient, and reliable operation by automating the elevator's movement between floors. Ladder diagrams are used due to their simplicity, graphical nature, and resemblance to electrical relay logic. This makes the design process easier for both programming and troubleshooting.

The proposed system operates based on inputs such as floor call buttons, door sensors, and position detectors. These inputs are processed by the PLC to generate output commands for motors, indicators, and door mechanisms. The control logic ensures that the elevator moves in the correct direction, stops precisely at each floor, and opens or closes the doors safely. Proper interlocks are included to prevent unsafe operations like door opening during motion. This logical approach guarantees smooth and coordinated control of all elevator functions.

Safety and reliability are key aspects considered in the design of the elevator ladder diagram. Emergency stop functions, overload protection, and door interlocks are implemented to enhance user safety. The system also includes priority logic to handle multiple floor requests efficiently. By structuring the ladder program systematically, it becomes easy to identify and correct faults when necessary. This ensures that the elevator system remains dependable and user-friendly.

In conclusion, the ladder diagram for the elevator system demonstrates the practical application of PLCs in industrial automation. It highlights how ladder logic can effectively manage complex operations using simple visual programming. The design provides flexibility, scalability, and ease of modification for future expansion. Through this project, the advantages of PLC-based automation in improving efficiency, safety, and performance are clearly established. Hence, this work serves as a model for understanding automation in intelligent building systems.

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1. INTRODUCTION

An elevator system is an important part of modern infrastructure, providing a safe and convenient means of vertical transportation in multi-story buildings. With the growing demand for automation, elevators have evolved from simple mechanical devices to intelligent, automated systems. Automation ensures smoother operation, better safety, and higher energy efficiency. The use of modern control systems makes elevators more reliable and easier to maintain. Among various control methods, **Programmable Logic Controllers (PLCs)** have become the most preferred choice for automation.

A **PLC** is a microprocessor-based device used to control industrial and mechanical processes through logical programming. It offers flexibility, accuracy, and high reliability in operation, which makes it ideal for elevator control systems. The **ladder diagram**, one of the most common PLC programming methods, is used to represent control logic in a graphical form. It closely resembles traditional electrical relay circuits, making it simple to design and understand. Engineers can easily modify or troubleshoot the system by referring to this diagram.

In an elevator system, the ladder diagram defines the logic for key operations such as floor selection, movement direction, door control, and safety functions. Inputs like floor call buttons, sensors, and limit switches are processed by the PLC to control outputs such as motor operation and door mechanisms. The PLC ensures that the elevator moves smoothly between floors, stops accurately at the desired level, and opens or closes doors safely. Additionally, it includes interlocks to prevent unsafe conditions, such as the door opening while the elevator is in motion.

The use of ladder logic in elevator systems enhances efficiency, safety, and reliability while reducing human intervention. It also provides the flexibility to expand or modify the system as needed, making it suitable for both small and large-scale applications. By integrating PLCs and ladder diagrams, modern elevators achieve precise control and improved performance. This introduction establishes the importance of automation and ladder logic programming in developing intelligent and dependable elevator systems for present and future use.

2.PROBLEM STATEMENT

In modern multi-story buildings, elevators play a vital role in ensuring safe, quick, and convenient movement of people and goods. However, traditional elevator systems that rely on relay-based control circuits often face issues such as complex wiring, frequent maintenance, and difficulty in troubleshooting. These systems lack flexibility and scalability, making it challenging to modify or upgrade them according to building requirements. Therefore, there is a need for a more efficient and reliable control mechanism that can simplify design and improve overall system performance.

Another major problem is the absence of intelligent control in conventional systems, which can lead to delays, unsafe operations, and inefficient power usage. For example, improper door operation, inaccurate floor stopping, and poor coordination between input and output devices can cause safety hazards and user dissatisfaction. Without a centralized and programmable control logic, managing multiple floors and optimizing elevator movement becomes difficult. This results in reduced efficiency, longer waiting times, and a higher chance of mechanical or electrical failures.

To overcome these limitations, an automated control system using a **Programmable Logic Controller (PLC)** programmed with a **ladder diagram** is proposed. The ladder logic provides a simplified, modular, and easily modifiable solution for elevator control. It allows precise coordination between sensors, motors, and safety devices, ensuring reliable operation and enhanced passenger safety. By replacing traditional relay-based systems with PLC-based ladder logic, the design becomes more efficient, easier to troubleshoot, and adaptable to future expansions or technology upgrades.

3.OBJECTIVES

The primary objective of the **Ladder Diagram for an Elevator System** project is to design and implement an automated, reliable, and efficient elevator control system using a Programmable Logic Controller (PLC). The system aims to ensure smooth vertical transportation, accurate floor selection, and safe door operation while minimizing human intervention and operational errors. This project focuses on improving safety, flexibility, and efficiency in elevator operation through the use of ladder logic programming.

The specific objectives of the project are as follows:

1. **To design and develop** a PLC-based ladder diagram that controls elevator movement, floor selection, and door operation.

2. **To automate** the elevator functions using inputs from floor buttons, sensors, and limit switches for precise and efficient control.

3. **To ensure safety and reliability** by implementing interlocks, emergency stop controls, and overload protection mechanisms.

4. **To achieve smooth operation** of the elevator by maintaining proper coordination between motor control and sensor feedback.

5. **To reduce manual control and mechanical complexity** compared to traditional relay-based elevator systems.

6. **To develop a flexible and scalable system** that can easily accommodate additional floors or upgraded features.

7. **To demonstrate** the practical application of PLC programming and ladder logic in modern automation systems.

8. **To provide an easily maintainable and cost-effective** control system suitable for industrial, residential, and commercial use.

4.COMPONENTS

The Automated Bottle Filling System designed in Automation Studio consists of electrical, pneumatic, and control components that work together to simulate the automatic filling process. Each component has a specific function in controlling the movement of bottles, detecting their position, and filling them with the required amount of liquid. The main components used in the simulation are described below:

1. Programmable Logic Controller (PLC)

The PLC is the core control unit of the system. It receives signals from the sensors and controls the output devices such as solenoid valves and conveyor motors. The PLC logic is programmed using ladder diagrams to ensure automatic operation — starting, stopping, and sequencing the filling cycle without manual intervention.

2. Proximity Sensor

A proximity sensor is used to detect the presence of the bottle on the conveyor. When a bottle reaches the filling position, the sensor sends a signal to the PLC to stop the conveyor and activate the filling valve. In the simulation, this sensor acts as a digital input to the PLC.

3. Conveyor Motor

The conveyor motor is used to move the bottles along the conveyor belt. It is controlled by the PLC through an output relay. The motor starts or stops according to the signal received from the PLC, ensuring proper synchronization with the filling process.

4. Solenoid Valve

The solenoid valve controls the flow of liquid (or air in simulation) into the bottle. When activated by the PLC, it allows the liquid to flow for a predefined time, ensuring accurate filling levels. In the pneumatic circuit, it can be represented as a 5/2 or 3/2 solenoid-actuated valve.

5. Pneumatic Cylinder

A pneumatic cylinder can be used to simulate mechanical actions such as moving the filling nozzle up and down or operating bottle stoppers. The cylinder extends or retracts based on signals from the solenoid valve, converting pneumatic energy into linear motion.

6. Relay Module

The relay acts as an interface between the PLC and the higher-power components (such as the conveyor motor or solenoid valve). It isolates the control circuit from the power circuit, ensuring safe operation and signal amplification.

7. Power Supply

A DC or AC power supply provides the required voltage to the PLC, sensors, relays, and actuators. It ensures that all electrical and control components operate reliably during the simulation.

8. Start and Stop Push Buttons

These manual control switches are used to start or stop the system. They act as digital inputs to the PLC and are used to initialize or halt the automatic bottle filling process.

9. Indicator Lamps

Indicator lamps (LEDs) are used to display the system's status, such as "Power ON," "Bottle Detected," "Filling in Progress," or "Cycle Complete." These provide visual feedback during simulation.

10. Timer and Counter (PLC Functions)

The timer is used to control the filling duration, ensuring that the solenoid valve remains open only for a specified period. The counter is used to keep track of the number of bottles filled, which helps in monitoring production performance.

11. Connecting Elements and Wiring

In Automation Studio, electrical connectors, pneumatic lines, and signal wires are used to establish connections between input and output components, ensuring correct data and power flow throughout the system.

5.WORKING PRINCIPLE

The PLC-based elevator system operates on the principle of programmable automation using a **Programmable Logic Controller (PLC)**. The system is designed to control elevator movement, floor selection, and door operation automatically using inputs from floor buttons, sensors, and limit switches. The PLC executes the pre-programmed **ladder logic** to coordinate all elevator functions in a safe, efficient, and reliable manner without human intervention. The working can be explained step-by-step as follows:

1. **System Initialization**
   - When the **Start push button** is pressed, the system becomes active, and power is supplied to the PLC, motors, sensors, and other connected components.
   - The PLC executes the pre-programmed ladder logic, initializing all relays, timers, and actuators.
2. **Floor Selection**
   - Floor call buttons inside the elevator and on each floor send digital input signals to the PLC when pressed.
   - The PLC determines the direction of movement (up or down) based on the requested floor and current position of the elevator.
3. **Elevator Movement**
   - The PLC activates the elevator motor to move the car in the desired direction.
   - Limit switches and position sensors provide feedback to the PLC to ensure accurate stopping at the selected floor.
4. **Door Operation**
   - Once the elevator reaches the selected floor, the PLC stops the motor and sends a signal to open the doors.
   - Timers ensure that the doors remain open for a pre-set duration before closing automatically.
5. **Safety Interlocks**
   - Sensors and interlocks prevent the elevator from moving if the doors are open or if an emergency stop is pressed.
   - Overload sensors ensure the elevator does not operate beyond safe weight limits, preventing accidents.
6. **System Stop and Emergency Handling**
   - The **Stop push button** or emergency signal can halt the elevator immediately.
   - The PLC continuously monitors sensors and inputs to ensure safe operation, preventing collisions, door obstruction, or motor malfunction.

6. LADDER DIAGRAM

Input and Output of Ladder diagram

		Name	Data type	Address
1		first floor indication	Bool	%Q0.0
2		second floor indication	Bool	%Q0.1
3		ground floor indication	Bool	%Q0.2
4		motor up	Bool	%Q0.3
5		motor down	Bool	%Q0.4
6		door open	Bool	%Q0.5
7		door close	Bool	%Q0.6
8		first floor sw	Bool	%M0.0
9		second floor sw	Bool	%M0.1
10		ground floor sw	Bool	%M0.2
11		second to ground	Bool	%M0.3
12		ground to second	Bool	%M0.4
13		door button	Bool	%M0.5

7.APPLICATIONS

1. **Residential Buildings:**
Used in apartments and housing complexes for safe and convenient vertical movement between floors.

2. **Commercial Buildings:**
Employed in offices, shopping malls, and hotels to handle high passenger traffic efficiently.

3. **Industrial Facilities:**
Utilized in factories and warehouses for transporting materials and equipment between different levels.

4. **Hospitals and Healthcare Centers:**
Designed to ensure smooth and reliable movement of patients, medical staff, and equipment.

5. **Educational Institutions:**
Installed in schools, colleges, and universities for accessibility and ease of movement.

6. **Parking Systems:**
Used in multi-level parking structures to move vehicles or platforms between floors automatically.

7. **High-Rise and Smart Buildings:**
Integrated with building automation systems for intelligent control and energy-efficient operation.

8. **Goods Elevators (Lifts):**
Applied in manufacturing and storage facilities for vertical transport of heavy loads or materials.

9. **Public Infrastructure:**
Used in airports, metro stations, and government buildings for passenger convenience and accessibility.

10. **Research and Training:**
Serves as a practical educational project to understand **PLC programming, ladder logic**, and automation concepts.

8.ADVANTAGES

1. **Reliable Operation**

- PLC-based ladder diagrams provide precise and consistent control of elevator movement, ensuring smooth operation and accurate floor alignment.

2. **Enhanced Safety**

- Safety features such as emergency stops, door interlocks, and overload protection help prevent accidents and protect passengers**.**

3. **Ease of Troubleshooting**

- The graphical representation of ladder logic makes it easy to understand the system, identify faults, and perform maintenance efficiently.

4. **Flexibility and Scalability**

- The system can be easily modified to accommodate additional floors, upgraded features, or changes in operational requirements**.**

5. **Reduced Human Intervention**

- Automation minimizes manual control, reducing human error and improving overall system efficiency.

6. **Cost-Effective Maintenance**

- Over the long term, PLC-based elevator systems require less maintenance than traditional relay-based systems, lowering operational costs.

9.DISADVANTAGE

1. **High Initial Cost**

- Implementing a PLC-based elevator system requires investment in hardware such as PLCs, sensors, actuators, motors, and supporting electrical infrastructure. This can be expensive for small-scale buildings or projects.

2. **Technical Complexity**

- The system requires knowledge of PLC programming, ladder logic design, and sensor integration, which can be challenging for operators or maintenance personnel without proper training.

3. **Limited Flexibility**

- Modifications such as adding more floors, upgrading features, or changing elevator capacity often require reprogramming the PLC and adjusting sensors, which can be time-consuming.

4. **Dependence on Sensors and Control Systems**

- Faulty sensors, misaligned switches, or programming errors can disrupt elevator operation. Regular maintenance and calibration are necessary to ensure reliable performance.

5. **Power Dependency**

- The system relies entirely on electrical power, and any power outage or electrical fault can stop elevator operation unless backup power systems are installed.

6. **Maintenance Requirements**

- Although automation reduces manual control, the PLC, motors, sensors, and other components require regular maintenance to ensure safety and long-term reliability.

10.CONCLUSION

The ladder diagram for an elevator system demonstrates the effective application of **PLC-based automation** in modern vertical transportation. By converting inputs from floor buttons, sensors, and switches into precise control of motors, doors, and indicators, the system ensures accurate, reliable, and efficient operation. This approach eliminates many of the problems associated with traditional relay-based systems, such as complex wiring, frequent maintenance, and limited scalability.

The project highlights the importance of safety and efficiency in elevator operation. With features like emergency stops, overload protection, and door interlocks, the system prioritizes passenger safety while maintaining smooth and coordinated movement between floors. Additionally, automation reduces human intervention, minimizes errors, and improves overall reliability, making the system suitable for residential, commercial, and industrial applications.

Finally, implementing ladder logic in PLCs provides flexibility for future modifications and expansions. The system can be easily adapted to add more floors, incorporate advanced control features, or integrate with smart building management systems. Overall, this project demonstrates how PLC-based elevator control systems combine **efficiency, safety, and scalability**, serving as a practical example of automation in real-world applications.

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